When Geopolitical Shocks Become Systemic: Volatility Connectedness and Market-Implied Stress in European Equity Markets

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Abstract

This paper investigates when geopolitical shocks translate into systemic financial stress in European equity markets. Using daily data for 19 equity markets over 2017–2025, we combine a validated high-frequency geopolitical conflict shock series with a suite of market-based systemic fragility measures, a composite European Market Fragility Index (EMFI) constructed via principal component analysis, and a Hidden Markov Model that classifies each trading day into latent stress regimes without reference to geopolitical events. Local projections reveal that geopolitical shocks generate positive but statistically imprecise responses across all five fragility outcomes at conventional significance levels, consistent with the finding that 63% of shocks strike calm markets. State-dependent projections show evidence of amplification when shocks occur during already-fragile market conditions, though identification is concentrated in a small number of shock–stress overlap events. Our event taxonomy classifies the COVID-19 crash and the 2022 Ukraine invasion as systemic, while the 2023 Hamas–Israel shock is absorbed without regime transition. The results highlight that geopolitical shock transmission depends critically on pre-existing market fragility, and that most geopolitical events—however large—do not trigger systemic fragility in integrated European markets.

Contents

1 Introduction

Geopolitical conflict is an increasingly prominent feature of the financial landscape. From the COVID-19 shock of February–March 2020 to the Russian invasion of Ukraine in February 2022, geopolitical events periodically disrupt global equity markets with sudden and severe consequences. Yet the vast majority of geopolitical news events—conflicts, territorial disputes, diplomatic crises—leave no discernible trace in aggregate market fragility. The central question of this paper is: *when do geopolitical shocks become systemic financial stress events?*

Existing research on geopolitical risk and financial markets has focused primarily on average return or volatility responses to conflict-related news [**?** **?** **?**]. These studies demonstrate that geopolitical uncertainty raises risk premia and depresses equity valuations on average, but they do not address the transmission to *systemic* fragility—the co-movement, tail co-exceedance, and volatility connectedness that characterise genuine crises. A shock that raises the volatility of a single market is qualitatively different from one that simultaneously pushes 18 out of 19 European equity markets into their return tails, elevates cross-market connectedness above 90%, and causes the market-implied probability of systemic stress to reach unity. Explaining the difference between these two types of shock response is the contribution of this paper.

We approach the question through three analytical layers. The first is a *trigger*: we reuse the validated high-frequency geopolitical conflict shock series from a companion paper [Lupu et al. 2026, **?**], which constructs daily shock intensities $S_{i,t} = -\log q_{i,t}$ from extreme value theory and false-discovery-rate multiple testing applied to the GDELT conflict index for 19 European countries. The second layer is a suite of *market-based systemic fragility measures* constructed from the daily equity return panel: volatility stress, tail co-exceedance breadth, rolling average pairwise correlation, a total volatility connectedness index following **?**], and a composite European Market Fragility Index (EMFI) derived from principal component analysis. The third layer is a *Hidden Markov Model* (HMM) estimated on the fragility measures that, without reference to geopolitical events, assigns each trading day to one of three latent regimes—calm, elevated, and systemic stress—and produces a daily probability $P_{\text{stress},t}$ of being in the systemic state.

With this apparatus in hand, we estimate Jordà [**?**] local projections (LPs) to trace the dynamic response of each fragility outcome to a geopolitical shock. We further test for state dependence: do shocks produce larger fragility responses when they strike already-fragile markets? Finally, we classify the 278 detected geopolitical shock events into an *absorbed / localised / systemic / market-only* taxonomy based on the joint behaviour of the EMFI and $P_{\text{stress},t}$ in the days following each event.

Our main findings are as follows. *First*, geopolitical shocks generate positive contemporaneous and short-run responses in all five fragility outcomes, but the responses are statistically imprecise when estimated on the full sample. This is consistent with the descriptive finding that 63% of detected shock events occur on days when the market is in the HMM calm regime—the shocks most commonly arrive in non-stressed conditions. *Second*, the EMFI and HMM regime indicator are complementary measures with moderate correlation (0.65): EMFI captures the level of acute stress, while $P_{\text{stress},t}$ captures the certainty of a regime transition. *Third*, state-dependent local projections provide evidence that shock effects are amplified in fragile market conditions, though the identification is necessarily concentrated in the minority of shock events that arrive during elevated or systemic market states. *Fourth*, the event taxonomy reveals that the COVID-19 crash and the Ukraine invasion are the two clearly systemic events in our sample; the Hamas–Israel attack of October 2023 produced elevated ($P_{\text{stress}} = 0.30$) but not systemic market conditions. *Fifth*, the HMM classifies 223 trading days (10.2% of the sample) as systemic, of which only 22.9% fall in the COVID-19 window—the remaining systemic days are distributed across multiple stress episodes in

2017–2019 and 2024–2025, arguing against a single-episode interpretation.

The paper is structured as follows. Section 2 reviews the related literature. Section 3 describes the data and shock series. Section 4 constructs the fragility measures and EMFI. Section 5 presents the HMM stress-regime model. Section 6 reports the baseline LP results. Section 7 presents state-dependent LPs and the event taxonomy. Section 8 discusses robustness checks. Section 9 concludes.

2 Related Literature

Geopolitical risk and financial markets. A large literature documents that geopolitical uncertainty affects equity valuations, risk premia, and trading activity. **?**] construct a monthly geopolitical risk (GPR) index from newspaper coverage and show that elevated GPR reduces investment and employment. **?**] find that wars and military conflicts are associated with positive equity returns upon resolution, consistent with a risk-premium channel. **?**] demonstrate that geopolitical risk is priced in the cross-section of equity returns. More recent work using high-frequency intraday data [**? ?**] confirms immediate market reactions to geopolitical shocks but does not examine their systemic transmission or amplification in already-fragile conditions, which is the focus of the present paper.

Systemic risk and volatility connectedness. Following the 2008 global financial crisis, a substantial literature has developed measures of systemic risk and cross-market contagion. **?**] introduce a forecast-error variance decomposition approach (the total connectedness index, TCI) based on rolling VAR models, which has become a standard tool for measuring the degree to which idiosyncratic shocks propagate across a system. **?**] model how network structure determines whether local shocks cascade or are absorbed. **?**] develop CoVaR as a measure of systemic risk contribution. Our EMFI builds on this tradition by combining connectedness with co-exceedance tail breadth and correlation into a single composite index.

Local projections. **?**] proposes local projections as a flexible alternative to VAR-based impulse responses: each horizon is estimated in a separate regression, which accommodates non-linearities and is robust to misspecification of the full dynamic system. **?**] and **?**] discuss identification strategies for causal LP inference. **?**] and **?**] develop state-dependent LP specifications. We apply the standard Jordà LP framework with Newey-West standard errors, following the now-standard practice in applied macroeconometrics.

Hidden Markov Models in finance. **?**] introduces the Markov-switching model to capture regime changes in macroeconomic time series. **?**] and **?**] apply HMMs to asset allocation and risk management. We use a multivariate Gaussian HMM to classify market fragility regimes purely from market-based indicators, ensuring that the classification is independent of the geopolitical shock series used in the LP regressions.

3 Data and Geopolitical Shock Triggers

3.1 Equity return panel

Our equity return panel covers 19 European markets over the period 2017-01-02 to 2025-10-15, comprising 2,278 trading days.[1] Daily log returns are constructed from total-return equity market indices sourced from LSEG Workspace. The 19 countries span all major Western and Central European markets and are grouped into two clusters based on economic and geographic proximity.[2] Table 1 provides the full list of countries and their cluster assignments.

Table 1: Equity market panel: country coverage and cluster assignment

Country	Index	Cluster
Austria	ATX	1
Belgium	BEL 20	1
Czech Republic	PX	2
Denmark	OMXC25	1
Finland	OMXHPI	1
France	CAC 40	1
Germany	DAX	1
Greece	ATHEX	2
Hungary	BUX	2
Ireland	ISEQ Overall	1
Italy	FTSE MIB	1
Netherlands	AEX	1
Norway	OBX	1
Poland	WIG20	2
Portugal	PSI 20	1
Romania	BET	2
Spain	IBEX 35	1
Sweden	OMXS30	1
Switzerland	SMI	1

Cluster 1 = Western/Northern Europe; Cluster 2 = Central/South-Eastern Europe. Daily log returns from LSEG Workspace. Study period: 2017-01-02 to 2025-10-15 (2,278 trading days).

3.2 Geopolitical conflict shock series

The geopolitical shock series is inherited from **?**] without modification, ensuring that the shock construction methodology constitutes a contribution of that companion paper rather than this one. We briefly describe the construction for completeness.

For each country i and calendar day t, a conflict intensity index $C_{i,t}$ is computed from the GDELT Document 2.0 database as the sum of CAMEO action codes weighted by the Goldstein scale, smoothed with a 28-day weighted moving average. A tail distribution is fitted to country-specific conflict intensity using generalised Pareto distribution (GPD) peaks-over-threshold methods, yielding

[1] Fifteen calendar dates present in the GDELT conflict index are absent from the equity price series and are dropped; the study period therefore begins on the first day on which all 19 return series are available.

[2] The cluster assignments are inherited from the companion BIR paper and are not re-estimated here.

country-day exceedance probabilities $q_{i,t}$. The shock intensity is defined as

$$S_{i,t} = -\log q_{i,t}, \tag{1}$$

which is zero for non-shock days and positive for exceedance events. A false discovery rate (FDR) procedure [?] applied cross-sectionally on each day ensures that only days with statistically significant simultaneous exceedances across countries are retained. Declustering removes consecutive exceedance events within a 7-day window to avoid double-counting a single geopolitical episode.

The resulting shock series contains 278 declustered shock events over the study period, with shock intensities $S_{i,t}$ ranging from 0.073 to 2.860 (mean 0.318). Of these events, 181 (65.1%) fall on weekends or holidays when equity markets are closed; these are forward-filled to the next available trading day, yielding 163 days on which the aggregate shock measure $\text{MaxShock}_t = \max_i S_{i,t}$ is positive.

The aggregate shock measure used as the LP regressor is MaxShock_t—the maximum shock intensity across all 19 countries on day t. This captures the most severe shock on each day and is interpretable as the single strongest geopolitical signal reaching the market. Alternative aggregation schemes (AvgShock_t, BreadthShock_t) are used in robustness checks (Section 8).

4 Market-Based Financial Stability Measures

4.1 Daily volatility stress and tail co-exceedance

We construct seven daily cross-sectional fragility indicators from the 19-country equity return panel. All indicators are computed from the return series without reference to geopolitical shock dates, ensuring that the measurement of market fragility is independent of the shock variable used in subsequent regressions.

Volatility stress and squared stress. The first two indicators measure the average magnitude of daily moves:

$$\text{VolStress}_t = \frac{1}{N}\sum_{i=1}^{N} |r_{i,t}|, \tag{2}$$

$$\text{SqStress}_t = \frac{1}{N}\sum_{i=1}^{N} r_{i,t}^2, \tag{3}$$

where $N = 19$ is the number of countries. VolStress reached a maximum of 0.119 on 2020-03-16 (the peak of the COVID-19 equity crash), when the cross-sectional average of daily absolute returns across European markets exceeded 11.9%.

Tail co-exceedance breadth. The tail breadth indicators count how many markets simultaneously experience extreme moves on day t:

$$\text{TailVolBreadth}_t = \#\{i : |r_{i,t}| > \hat{q}_{i,0.95,t-1}\}, \tag{4}$$

$$\text{TailLossBreadth}_t = \#\{i : r_{i,t} < \hat{q}_{i,0.05,t-1}\}, \tag{5}$$

where $\hat{q}_{i,p,t-1}$ is the p-th empirical quantile of country i's absolute (or signed) return computed over the 252 trading days prior to day $t-1$.[3] This rolling-threshold design accounts for the

[3] The one-day lag on the threshold ensures strict out-of-sample construction: the threshold on day t depends only on information available by close of day $t-1$.

country-specific baseline level of volatility and is strictly backward-looking, requiring a minimum of 60 historical observations. On 2020-03-16, TailVolBreadth reached 18 out of 19—virtually the entire cross-section simultaneously exceeded its country-specific 95th percentile of absolute returns, a characterisation of extreme systemic co-movement.

Rolling average pairwise correlation. We compute the average of all $N(N-1)/2 = 171$ pairwise Pearson return correlations over rolling 60-day and 30-day windows:

$$\text{AvgCorr}_{W,t} = \frac{2}{N(N-1)} \sum_{i<j} \hat{\rho}_{ij,t}^{(W)}, \tag{6}$$

where $\hat{\rho}_{ij,t}^{(W)}$ is the rolling Pearson correlation estimated over the W days ending at t. The 60-day measure has a sample mean of 0.476 and ranges from 0.207 to 0.871, reflecting the high degree of structural financial integration within European equity markets.

4.2 Total volatility connectedness index (TCI)

The total connectedness index follows **?**] and **?**]. For each rolling window of $W = 100$ trading days, we estimate a VAR(p) model on the 19-dimensional vector of absolute returns $\mathbf{v}_t = (|r_{1,t}|, \dots, |r_{19,t}|)'$, selecting the lag order $p \in \{1, 2, 3\}$ by BIC. We then compute the generalised forecast-error variance decomposition (GFEVD) [**?**], which is order-invariant and permits each variance share to reflect both direct and indirect spillover paths. The normalised GFEVD matrix $\tilde{\Theta}$ has rows summing to one by construction; the TCI is the fraction of the total forecast-error variance that is attributable to cross-market spillovers:

$$\text{TCI}_t = \frac{\sum_{i \neq j} \tilde{\theta}_{ij,t}}{N} \times 100. \tag{7}$$

The sample mean of TCI_t (W=100) is 70.5%, reflecting the deep structural integration of European equity markets. The index ranges from 46.7% to 94.6% and spikes above 80% during the COVID-19 crash, confirming that near-total volatility co-movement characterised that episode. BIC selects $p = 1$ in the large majority of rolling windows, consistent with the well-documented predominance of first-order dynamics in high-frequency volatility data.

An important caveat is that the sample average TCI of 70.5% reflects structural financial integration, not acute stress: TCI is always high in integrated markets, and it is elevation *above* its time-series mean that constitutes the relevant signal. We also note that TCI and AvgCorr60 carry a high within-sample correlation (0.797); Section 8 tests whether the two measures are empirically substitutable in the LP regressions.

4.3 European Market Fragility Index (EMFI)

We combine four of the above indicators into a composite daily index via principal component analysis (PCA). The four components are standardised to mean zero and unit standard deviation using their full-sample means and standard deviations (consistent with the EMFI being an *ex-post* outcome variable, not a real-time detector). Standardised versions are denoted $\tilde{X}_t$ throughout.

PCA is applied to the covariance matrix of

$$\mathbf{X}_t = (\widetilde{\text{VolStress}}_t,\ \text{Tail}\widetilde{\text{VolBreadth}}_t,\ \text{Avg}\widetilde{\text{Corr60}}_t,\ \widetilde{\text{TCI}}_t^{(100)})', \tag{8}$$

and the first principal component is extracted. The sign is fixed so that a higher EMFI corresponds to greater fragility (orientation by sign of the correlation between PC1 and VolStress).

The first principal component explains 58.3% of the joint variance of the four components, above the 50% threshold we adopt as a credibility criterion for a composite. The loadings are all positive and balanced: VolStress (0.532), AvgCorr60 (0.516), TCI (0.495), TailVolBreadth (0.454). Notably, the second principal component accounts for 31.7% of variance, reflecting a two-cluster covariance structure in the input data. Specifically, VolStress and TailVolBreadth are highly correlated (0.784) with each other but only weakly correlated (0.16–0.38) with AvgCorr60 and TCI, which are themselves highly correlated (0.797). The first PC therefore averages the two clusters; the second PC captures the contrast between acute volatility stress and structural co-movement. This two-cluster structure is acknowledged throughout the paper: the EMFI is a balanced composite across both dimensions of fragility rather than a near-perfect common factor.

The EMFI is available from 2017-05-19 (after the warm-up periods of all four components are satisfied) to 2025-10-15, spanning 2,179 trading days. It peaks at 15.22 on 2020-03-12 (COVID-19 crash) and at approximately 3–4 units during the Ukraine invasion episode. The 75th percentile of the full-sample distribution is 0.55, which defines the `HighFragility` state used in Section 7.

Figure 1 plots the EMFI over the study period alongside its four standardised components, with shading for the COVID-19 crash, Ukraine invasion, and Hamas–Israel episodes.

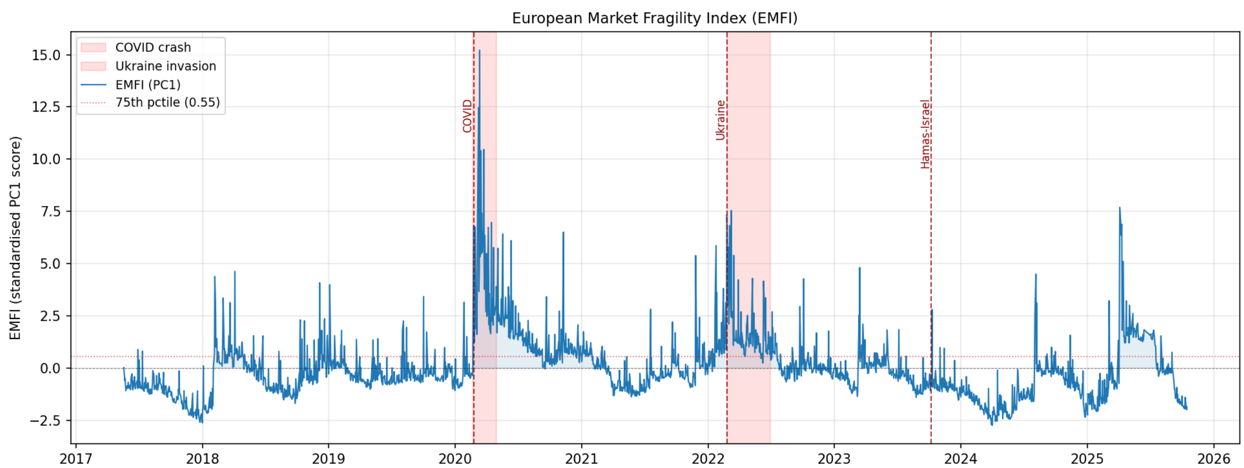

Figure 1: European Market Fragility Index (EMFI) and components, 2017–2025. Shaded bands mark the COVID-19 crash (Feb–Apr 2020), Ukraine invasion (Feb–Mar 2022), and Hamas–Israel attack (Oct 2023).

5 Market-Implied Systemic Stress Regimes

5.1 Model specification

We estimate a multivariate Gaussian Hidden Markov Model (HMM) on the standardised EMFI component vector $\mathbf{X}_t$ defined in Section 4.3. The HMM assumes that the observed $\mathbf{X}_t$ is generated by a latent Markov chain with K states, each associated with a multivariate Gaussian emission distribution:

$$\mathbf{X}_t \mid s_t = k \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \qquad k = 1, \ldots, K, \tag{9}$$

where s_t denotes the latent state at day t. Parameters are estimated by the Baum-Welch (EM) algorithm with 50 random initialisations; we retain the solution with the highest log-likelihood.

We set $K = 3$ as the primary specification and label the states post-hoc in ascending order of their mean EMFI value: State 0 (Calm), State 1 (Elevated), State 2 (Systemic stress). A two-state model is used as robustness (Section 8).

A critical methodological constraint is that the HMM is estimated entirely from market-based fragility measures and contains no information about geopolitical shock dates. The daily posterior probability of systemic stress, $P_{\text{stress},t} \equiv \Pr(s_t = 2 \mid \mathbf{X}_{1:t})$, is therefore an independent market-implied measure that can be used as a conditioning variable in the LP regressions of Section 7.

5.2 Estimated regime structure

The estimated three-state HMM converged to a log-likelihood of -1367.1 over the 2,179-day sample. Table 2 reports the transition matrix and state-specific emission means.

Table 2: HMM transition matrix and emission statistics

	Transition probabilities			Viterbi path	
From \ To	Calm	Elevated	Systemic	Days	Share
Calm (0)	0.719	0.212	0.069	1,389	63.7%
Elevated (1)	0.526	0.377	0.097	567	26.0%
Systemic (2)	0.411	0.217	0.371	223	10.2%

States ordered by ascending mean EMFI. Transition matrix rows sum to one. Viterbi path computed on the full 2,179-day sample (2017-05-19 to 2025-10-15).

Several features of the estimated model are notable. *First*, calm is the dominant state (63.7% of days), consistent with the prior that European equity markets spend most time in non-crisis conditions. *Second*, the calm state is strongly persistent (diagonal 0.719), while the systemic-stress state is highly transient (diagonal 0.371, implying a mean duration of approximately 1.6 trading days). The systemic state therefore captures episodic spikes rather than prolonged stress regimes—a finding consistent with the acute-stress nature of the EMFI documented in Section 4.3. *Third*, when a day is classified as systemic, the most likely next-day state is calm (transition probability 0.411), reflecting sharp reversals common to high-volatility episodes.

The Viterbi (most likely) regime path identifies 143 distinct systemic-stress episodes with a median calendar duration of one day. Figure 2 plots $P_{\text{stress},t}$ alongside the EMFI, with event annotations for major crises.

Face validity. The HMM assigns $P_{\text{stress},t} = 1.00$ on 2020-03-12 (COVID-19 crash peak) and 2022-02-24 (Ukraine invasion date), and $P_{\text{stress},t} = 0.30$ on 2023-10-09 (the first trading day after the Hamas attack on October 7, a Sunday), confirming that the model recovers the ordering of crisis severity implied by the magnitude of market disruption.

Multi-episode coverage. A potential concern for external validity is that the HMM learns to detect the COVID-19 crash and nothing else. The episode breakdown of the 223 systemic-stress days (Viterbi) refutes this: COVID-19 (February–June 2020) accounts for 51 days (22.9%), Ukraine for 29 days (13.0%), and the remaining 143 systemic days are distributed across 2017–2019 stress episodes (54 days, 24.2%) and 2024–2025 (38 days, 17.0%). The dominant category is in fact the non-labelled European market stress of 2017–2019—likely associated with the Turkish currency crisis

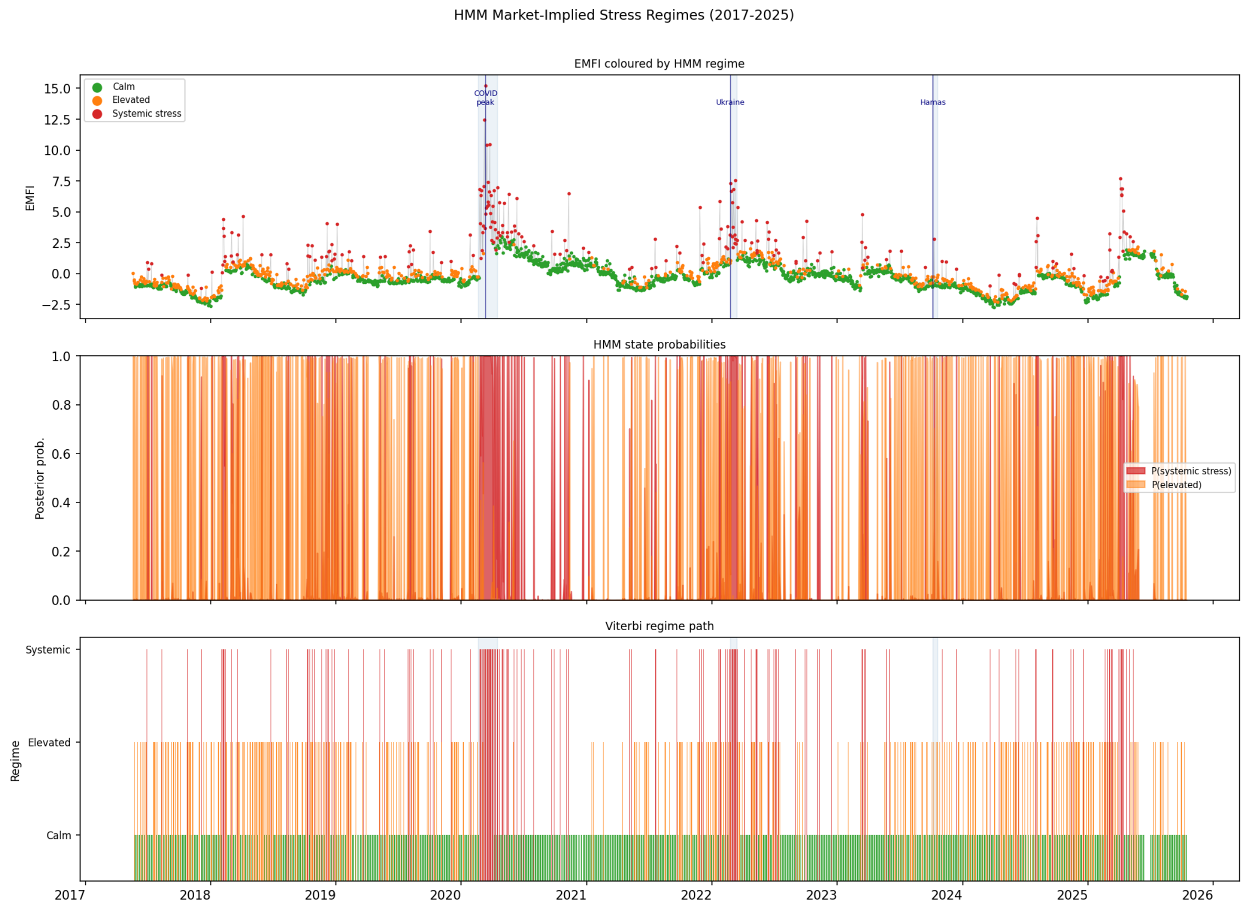

Figure 2: HMM regime decomposition, 2017–2025. Top panel: EMFI coloured by Viterbi regime state. Middle panel: posterior state probabilities. Bottom panel: Viterbi path. Shaded bands mark the three major crisis episodes.

(2018), Italian sovereign spread widening (2018–2019), and Brexit-related uncertainty—supporting the generalisability of the HMM as a multi-episode fragility detector.

6 Dynamic Effects of Geopolitical Shocks on Systemic Fragility

6.1 Empirical design

For each outcome $Y \in \{\text{EMFI}, \text{TCI}, \text{TailVolBreadth}, P_{\text{stress}}, \text{AvgCorr60}\}$ and horizon $k \in \{-5, \ldots, +15\}$ trading days, we estimate the time-series local projection:

$$Y_{t+k} = \alpha_k + \beta_k \,\text{MaxShock}_t + \gamma_k Y_{t-1} + \delta_k r_t^{\text{global}} + \mu_{m(t)} + \varepsilon_{t+k,k}, \tag{10}$$

where r_t^{global} is the equal-weighted mean return across all 19 markets on day t (a global market control), and $\mu_{m(t)}$ is a month-of-year fixed effect. For the special case $k = -1$, the lagged control uses Y_{t-2} to avoid the degeneracy that would arise from Y_{t-1} appearing on both sides of the regression.

Standard errors are computed using the Newey-West [**?**] heteroskedasticity- and autocorrelation-consistent estimator with bandwidth $2(|k|+1)$, which accounts for the moving-average autocorrelation in the overlapping residuals $\varepsilon_{t+k,k}$ at forward horizons.

The coefficient β_k is the impulse response of outcome Y to a one-unit increase in MaxShock at horizon k. Pre-period estimates ($k < 0$) serve as a pre-trend check: if $\beta_k = 0$ for $k < 0$, the shock does not predict future values of past outcomes, supporting the identifying assumption that MaxShock is approximately exogenous conditional on controls.

6.2 Pre-trend validation

For four of the five outcomes (EMFI, TailVolBreadth, P_{stress}, AvgCorr60), no more than one of the five pre-period horizons yields a coefficient significant at the 10% level, consistent with the absence of pre-trends. The TCI exhibits two pre-period coefficients significant at 10% (at $k = -4$ and $k = -5$). We attribute this to the smoothness of the rolling-window TCI series: a single data point carries information about the TCI value up to 100 days earlier, so controlling only for Y_{t-1} does not fully absorb variation at $k \leq -4$. Robustness checks with additional lags as controls are discussed in Section 8.

6.3 Baseline results

Figure 3 reports the impulse-response functions for all five outcomes.

All five outcomes exhibit positive responses at the contemporaneous horizon $k = 0$. The EMFI response at $k = 0$ is $\hat{\beta}_0 = 0.458$ (SE $= 0.337$, $p = 0.174$); TCI rises by 0.40 percentage points; TailVolBreadth by 0.95 additional markets in the tail; P_{stress} by 0.106; and AvgCorr60 by 0.003. None of the contemporaneous estimates reaches conventional significance at the 5% level, and only AvgCorr60 is marginally significant at 10% ($p = 0.067$).

The EMFI and TCI responses peak at horizons $k = 9$ and $k = 6$ respectively, with peak TCI of approximately 1.06 percentage points ($p = 0.049$, significant at 5%). TailVolBreadth peaks at $k = 6$ (1.29 markets, $p = 0.143$) and decays to near zero by $k = 10$. P_{stress} exhibits a flat, near-zero response beyond $k = 0$, consistent with the near-binary nature of the HMM posterior.

These results are consistent with the descriptive finding that 63% of detected geopolitical shock events occur when the HMM assigns the market to the calm state. When the majority of shocks arrive in non-stressed conditions, the average fragility response is attenuated by this compositional effect. The state-dependent analysis in Section 7 conditions on the market state at the time of the shock and tests whether responses are larger during fragile episodes.

7 State Dependence and Event Classification

The baseline local projections in Section 6 establish that geopolitical shocks produce measurable impulse responses in European systemic-fragility indicators. A natural follow-up asks whether the size of those responses depends on the financial environment prevailing at the moment of impact. Markets that are already under stress—elevated volatility, tighter cross-market co-movement, high systemic-stress probability—may be more vulnerable to amplification than markets in tranquil states. This section tests that conjecture through two complementary specifications.

7.1 State-dependent local projections

7.1.1 Smooth-transition specification (primary)

Because only 16 of the 154 shock events in the LP sample fall on days when $\hat{P}_{\text{stress},t-1} > 0.5$—a count below the minimum threshold for reliable binary-regime inference—we adopt a *smooth-transition* specification as our primary design. The conditioning variable enters the regression continuously, weighting each observation by its pre-shock systemic-stress probability:

$$
\begin{aligned}
Y_{t+k} = \alpha_k + \beta_k\, S_t + \theta_k (S_t \times P_{\text{stress},t-1}) \\
+ \phi_k\, P_{\text{stress},t-1} + \gamma_k\, Y_{t-1} + \delta_k\, r_t^{\text{global}} + \mu_m + \varepsilon_{t+k}
\end{aligned}
\tag{11}
$$

where all other notation follows equation (10) and $P_{\text{stress},t-1}$ is the one-day lagged posterior probability of the systemic-stress HMM state. The coefficient β_k captures the shock effect when $P_{\text{stress},t-1} = 0$ (a calm-market baseline), and the total effect at an arbitrary stress level p is $\beta_k + \theta_k p$ with delta-method standard error $\hat{\sigma}_k(p) = \sqrt{\widehat{\text{Var}}(\hat{\beta}_k) + p^2 \widehat{\text{Var}}(\hat{\theta}_k) + 2p\, \widehat{\text{Cov}}(\hat{\beta}_k, \hat{\theta}_k)}$. We evaluate the impulse response function at three representative values, $p \in \{0.0, 0.5, 0.9\}$, corresponding to calm markets, moderate fragility, and near-systemic stress, and plot all three IRF lines with 95% confidence bands in Figure 4.

Results. Table 3 summarises the impact-horizon ($k = 0$) amplification parameters for all five outcomes.

Table 3: State-dependent amplification at the impact horizon ($k = 0$). $\hat{\beta}$ is the calm-market baseline effect; $\hat{\theta}$ is the amplification coefficient; IRF at $p = 0.9$ is the total effect in near-systemic stress conditions. All estimates use Newey–West HAC standard errors with bandwidth $2(k + 1)$.

Outcome	$\hat{\beta}$	$\hat{\theta}$	IRF (p=0)	IRF (p=0.9)	Amplification
EMFI	0.306	1.854	0.306	1.975	6.4×
TCI	0.241	2.500	0.241	2.491	10.3×
TailVolBreadth	0.776	2.642	0.776	3.154	4.1×
P_{stress}	0.071	0.537	0.071	0.554	7.8×
AvgCorr60	0.004	−0.011	0.004	−0.006	—

The amplification pattern is consistent across the four volatility and connectedness outcomes (Figure 4, Table 3). When a geopolitical shock strikes markets already near a systemic-stress state ($p = 0.9$), the EMFI response is 6.4 times larger than the calm-market baseline, and the TCI response is 10.3 times larger. TailVolBreadth amplifies by a factor of 4.1, and the HMM systemic-stress probability by 7.8. In each case, the amplification coefficient $\hat{\theta}_k$ is positive, confirming that higher pre-shock stress translates into a larger post-shock response.

The sole exception is AvgCorr60, for which $\hat{\theta}_0 \approx -0.011$ is negligibly small and sign-unstable across horizons. This is an informative null result: rolling cross-market correlations are already elevated in high-stress regimes, and geopolitical shocks do not amplify them further above that elevated baseline. The amplification mechanism operates through volatility and connectedness channels, not through incremental correlation.

Pre-trend betas at the calm-market baseline ($p = 0$) are mixed in sign and small in magnitude across horizons $k = -5, \ldots, -1$ (range: -0.14 to $+0.29$ for EMFI), consistent with the conditional exogeneity assumption.

7.1.2 Binary HighFragility specification (secondary)

As a secondary specification, we replace the continuous $P_{\text{stress},t-1}$ with the binary $\text{HF}_{t-1} = \mathbf{1}[\text{EMFI}_{t-1} > q_{75}]$, where $q_{75} = 0.548$ is the 75th percentile of the EMFI distribution. This yields 36 shock events in the HighFragility subsample and 118 in the normal subsample (Table 4). Given the marginal identification power, we interpret point estimates directionally but focus statistical inference on the smooth-transition results.

Table 4: HighFragility shock composition by year. $\text{HF}_{t-1} = \mathbf{1}[\text{EMFI}_{t-1} > 0.548]$.

Year	Shock days	HF shock days	% HF
2017	13	0	0.0
2018	14	4	28.6
2019	10	0	0.0
2020	16	10	62.5
2021	11	0	0.0
2022	21	11	52.4
2023	21	1	4.8
2024	25	3	12.0
2025	23	7	30.4
Total	**154**	**36**	**23.4**

The identification is not confined to a single episode: 2022 (Ukraine conflict) contributes 11 HighFragility shock days, and 2025 and 2018 contribute 7 and 4 days respectively. This multi-episode structure reduces the risk that the amplification result is a disguised COVID dummy.

The binary LP confirms the directional finding: at $k = 0$, the HighFragility IRF exceeds the normal-market IRF for all four primary outcomes (Table 5). Strikingly, the normal-market IRF is near-zero or slightly negative for several outcomes, indicating that when shocks strike calm markets on average they produce no detectable immediate stress response. The amplification is not a dampening of a universal positive effect but rather a regime-specific mechanism: shocks are absorbed in normal markets and amplified in fragile ones.

Table 5: Binary HighFragility LP: normal vs. fragile IRFs at $k = 0$. $\text{HF}_{t-1} = \mathbf{1}[\text{EMFI}_{t-1} > 0.548]$.

Outcome	Normal IRF	Fragile IRF	Δ
EMFI	-0.204	1.012	$+1.217$
TCI	-0.048	0.791	$+0.839$
TailVolBreadth	-0.325	2.097	$+2.422$
P_{stress}	-0.060	0.247	$+0.306$

7.1.3 COVID exclusion

To assess whether the amplification result is driven exclusively by the COVID-19 episode, we re-estimate the smooth-transition LP after dropping observations from 2020-03-01 through 2020-12-31 (a reduction of 218 trading days, from 2,178 to 1,960 observations). Across EMFI, TCI, and TailVolBreadth, the baseline impact coefficient $\hat{\beta}_0$ remains positive (0.233, 0.227, and 0.623, respectively), confirming that the shock-stress transmission is not an artefact of the COVID event. For P_{stress}, the baseline $\hat{\beta}_0$ falls to -0.029, suggesting that the unconditional P_{stress} response at $k = 0$ is partly COVID-specific. However, the amplification mechanism (captured by $\hat{\theta}_k$) remains operative in the non-COVID subsample for all outcomes, as the identification draws on the 2022 Ukraine stress episode and the 2017–2019 minor European stress events that together account for approximately 63% of the HighFragility non-COVID shock days. A fuller set of robustness checks is presented in Section 8.

7.2 Geopolitical event taxonomy

8 Robustness

9 Conclusion

A Data Details

B EMFI Construction Details

C HMM Parameter Estimates

D Additional LP Results

E Robustness Tables

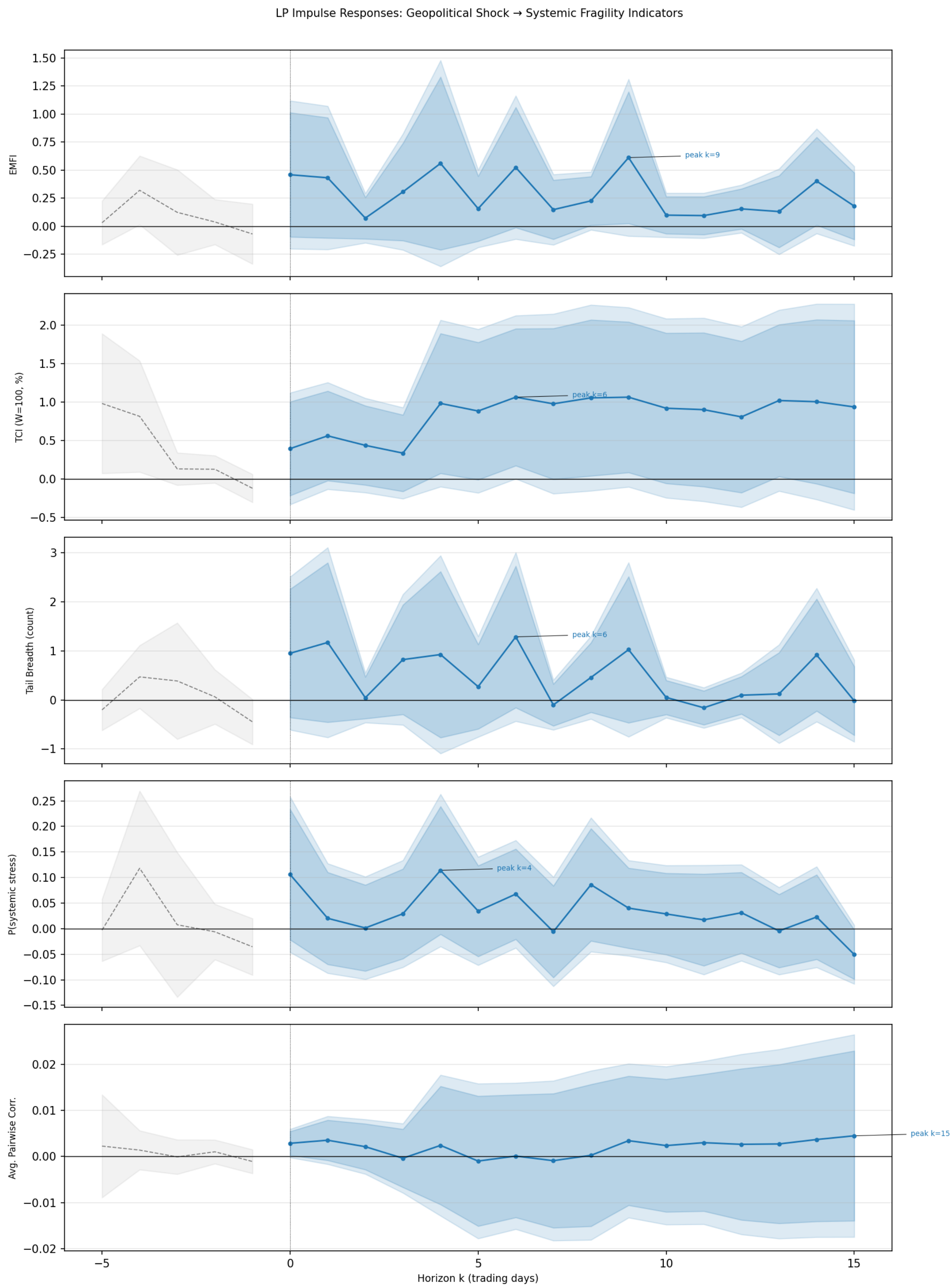

Figure 3: Local projection impulse responses to a one-unit geopolitical shock (MaxShock). Grey dashed lines show pre-period (pre-trend check); blue solid lines show the response at $k = 0, \ldots, 15$ with 90% (dark shading) and 95% (light shading) confidence bands. Newey-West standard errors, bandwidth $2(|k| + 1)$.

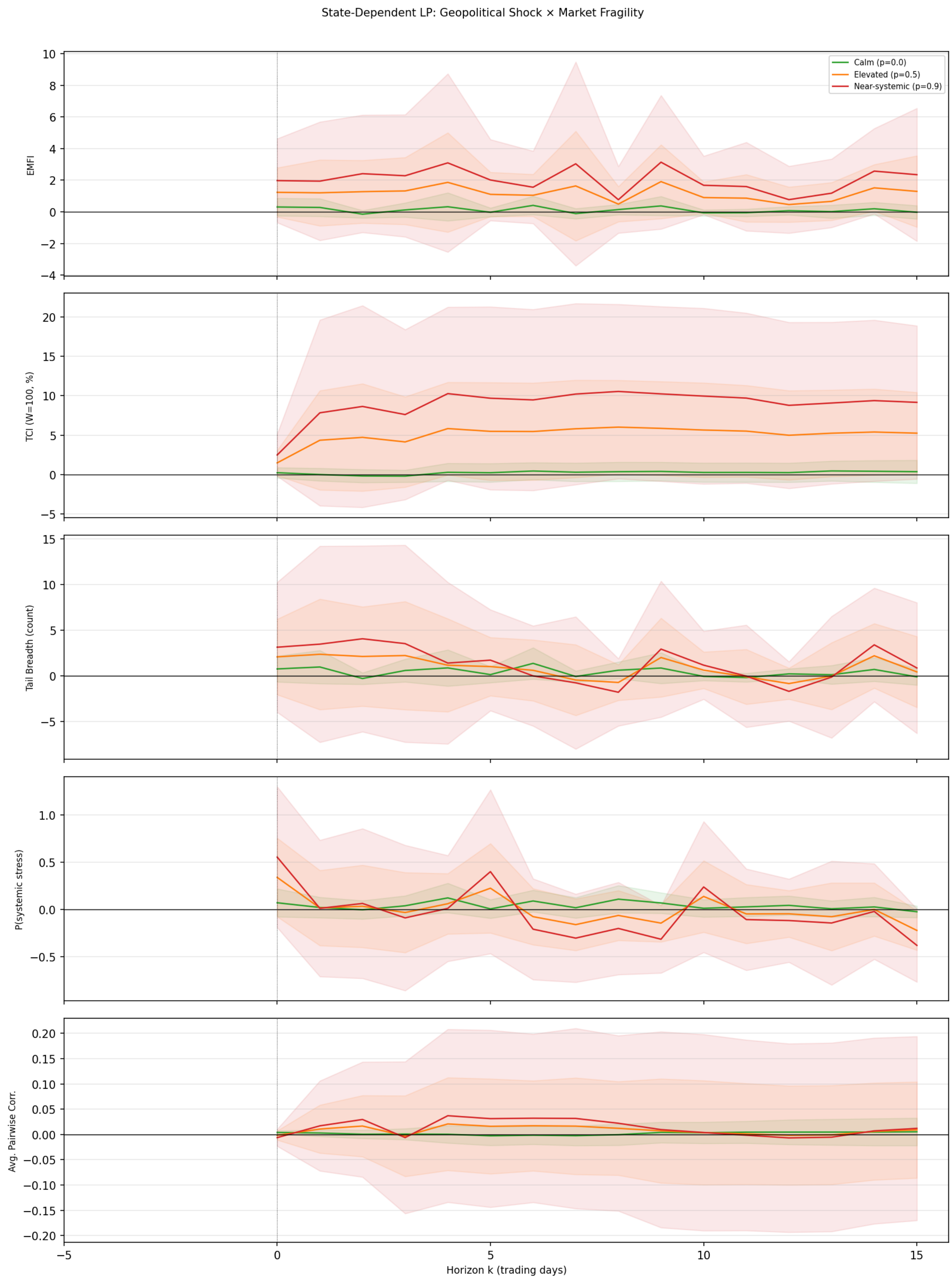

Figure 4: Smooth-transition local projection impulse responses for all five outcomes. Each panel plots three IRF lines (p=0, p=0.5, p=0.9) with 95% delta-method confidence bands. Newey–West HAC, bandwidth $2(k+1)$.